

SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
COMPUTER SCIENCE AND ENGINEERING

BRANCH : CSE
YEAR/SEM : II & III / IV & VI
SUB CODE & TITLE : CSA1322 - Theory of Computation
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Activity 1-4: DFA Design & Tracing

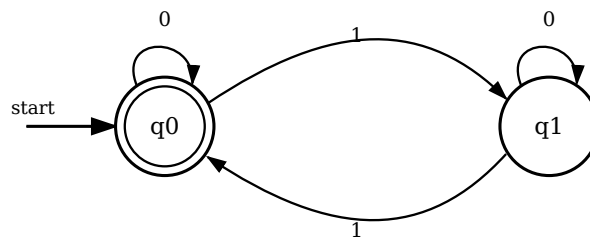
Activity 1: DFA Design - Even Number of 1s

Problem Statement: Design a Deterministic Finite Automaton (DFA) that accepts all binary strings over $\Sigma = \{0,1\}$ which contain an even number of 1s.

1. Formal Definition $M = (Q, \Sigma, \delta, q_0, F)$

- $Q = \{ q_0, q_1 \}$ — q_0 = "even 1s seen so far", q_1 = "odd 1s seen so far"
- $\Sigma = \{ 0, 1 \}$
- q_0 (start state) = q_0
- F (final/accepting states) = $\{ q_0 \}$ (zero is even, so the empty string is accepted)

2. State Transition Diagram



3. Transition Table

State	0	1	Type
$\rightarrow * q_0$	q_0	q_1	Start / Accepting
q_1	q_1	q_0	Rejecting

$\rightarrow$ = start state, $*$ = accepting state

4. Tracing the Given Inputs

- **1010:** $q_0 \xrightarrow{1} q_1 \xrightarrow{0} q_1 \xrightarrow{1} q_0 \xrightarrow{0} q_0$ (two 1s $\rightarrow$ even) $\rightarrow$ **ACCEPTED**
- **111:** $q_0 \xrightarrow{1} q_1 \xrightarrow{1} q_0 \xrightarrow{1} q_1$ (three 1s $\rightarrow$ odd) $\rightarrow$ **REJECTED**
- **1100:** $q_0 \xrightarrow{1} q_1 \xrightarrow{1} q_0 \xrightarrow{0} q_0 \xrightarrow{0} q_0$ (two 1s $\rightarrow$ even) $\rightarrow$ **ACCEPTED**

Language accepted: $L = \{ w \in \{0,1\}^* \mid w \text{ contains an even number of 1s} \}$

Activity 2: DFA for Strings Ending with "01"

Problem Statement: Construct a DFA over $\Sigma = \{0,1\}$ that accepts all strings that end with "01".

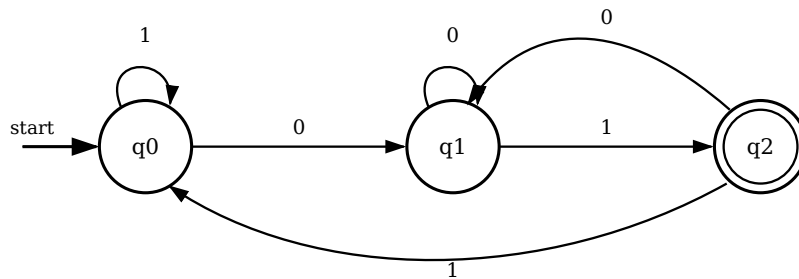
1. Formal Definition (minimum states)

- $Q = \{ q_0, q_1, q_2 \}$
- $\Sigma = \{ 0, 1 \}$
- q_0 = q_0 (no useful suffix seen yet)

- $F = \{ q_2 \}$ (last two symbols read were exactly "01")

Meaning of states: **q0** = last symbol read was not "0" (or nothing read); **q1** = last symbol read was "0"; **q2** = last two symbols read were "0" then "1" (i.e., string ends in "01").

2. State Transition Diagram



3. Transition Table

State	0	1	Type
→ q0	q1	q0	Start
q1	q1	q2	Intermediate
* q2	q1	q0	Accepting

4. Tracing the Given Inputs

- **01**: $q_0 \xrightarrow{0} q_1 \xrightarrow{1} q_2 \rightarrow \text{ACCEPTED}$
- **101**: $q_0 \xrightarrow{1} q_0 \xrightarrow{0} q_1 \xrightarrow{1} q_2 \rightarrow \text{ACCEPTED}$
- **1101**: $q_0 \xrightarrow{1} q_0 \xrightarrow{1} q_0 \xrightarrow{0} q_1 \xrightarrow{1} q_2 \rightarrow \text{ACCEPTED}$
- **100**: $q_0 \xrightarrow{1} q_0 \xrightarrow{0} q_1 \xrightarrow{0} q_1$ (ends in q1, not q2) → **REJECTED**

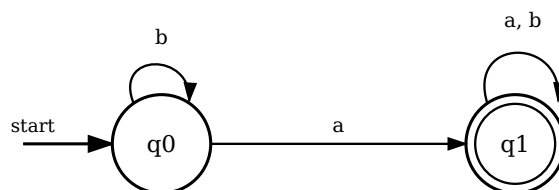
Language accepted: $L = \{ w \in \{0,1\}^* \mid w \text{ ends with the substring "01" } \}$

Activity 3: Identify Accept/Reject from Given DFA

Problem Statement: Given a DFA M that accepts strings over $\Sigma = \{a, b\}$ which contain at least one 'a'.

DFA Used

- $Q = \{ q_0, q_1 \}$ — q_0 = "no 'a' seen yet", q_1 = "at least one 'a' seen"
- $q_0 = q_0$ (start / rejecting state)
- $F = \{ q_1 \}$ (the accepting state)



Transition Table

State	a	b	Type
→ q0	q1	q0	Start / Rejecting
* q1	q1	q1	Accepting

Tracing the Given Inputs

- **bbbb:** $q_0 \xrightarrow{b} q_0 \xrightarrow{b} q_0 \xrightarrow{b} q_0 \xrightarrow{b} q_0$ (no 'a' encountered) → **REJECT**
- **abbb:** $q_0 \xrightarrow{a} q_1 \xrightarrow{b} q_1 \xrightarrow{b} q_1 \xrightarrow{b} q_1$ → **ACCEPT**
- **bab:** $q_0 \xrightarrow{b} q_0 \xrightarrow{a} q_1 \xrightarrow{b} q_1$ → **ACCEPT**
- **ϵ (empty string):** remains at q_0 (no input processed) → **REJECT**

Language accepted (in words): $L(M)$ is the set of all strings over $\{a, b\}$ that contain **at least one occurrence of the symbol 'a'**. Equivalently, it excludes only strings made entirely of b's (including the empty string).

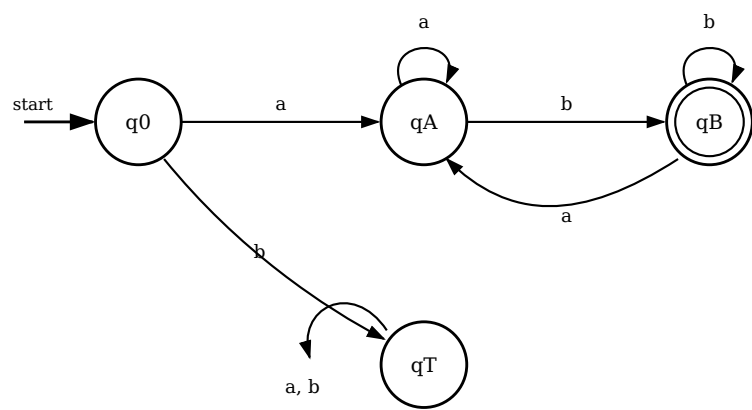
Activity 4: Real-Time Application DFA - Password Rule

Problem Statement: Design a DFA that accepts all strings over $\Sigma = \{a, b\}$ that (1) start with 'a' AND (2) end with 'b'.

1. Formal Definition

- $Q = \{ q_0, q_A, q_B, q_T \}$
- $\Sigma = \{ a, b \}$
- $q_0 = q_0$ (start, nothing read yet)
- $F = \{ q_B \}$ (string started with 'a' and currently ends with 'b')
- q_T = trap/dead state for any string that does not start with 'a'

2. State Transition Diagram



3. Transition Table

State	a	b	Type
$\rightarrow q_0$	q_A	q_T	Start
q_A	q_A	q_B	Intermediate (started with 'a')
$* q_B$	q_A	q_B	Accepting (starts 'a', ends 'b')
q_T	q_T	q_T	Trap / Rejecting

4. Checking the Given Strings

- **ab:** $q_0 \xrightarrow{a} q_A \xrightarrow{b} q_B$ → **ACCEPTED**
- **aab:** $q_0 \xrightarrow{a} q_A \xrightarrow{a} q_A \xrightarrow{b} q_B$ → **ACCEPTED**
- **abb:** $q_0 \xrightarrow{a} q_A \xrightarrow{b} q_B \xrightarrow{b} q_B$ → **ACCEPTED**
- **ba:** $q_0 \xrightarrow{b} q_T \xrightarrow{a} q_T$ (does not start with 'a') → **REJECTED**
- **a:** $q_0 \xrightarrow{a} q_A$ (starts with 'a' but does not end with 'b') → **REJECTED**

Language accepted: $L = \{ w \in \{a,b\}^* \mid w \text{ begins with 'a' AND ends with 'b' } \}$

Evaluation Rubric

Criteria	Marks
Correct DFA design	4

Correct final states	2
Correct tracing	2
Neat diagram & explanation	2
Total	10 Marks